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MOST FREQUENTLY REPEATED CONCEPTUAL AREAS IN BIOCHEMISTRY-II BIO-506T (TOPICS 1–80) MIDTERM EXAMS

Main Group (BS Biological Sciences) *https://chat.whatsapp.com/DWXwJpXahbxFVQutey8JXN*

BIO506 – Biochemistry II *https://chat.whatsapp.com/EMIThWXx70oKoMWlhdeiL2*

Q1. The energy available to perform useful work at constant temperature and pressure is called _____.

Answer: Free energy

Q2. The change in free energy is represented by _____.

Answer: ΔG

Q3. A reaction with a negative ΔG is considered _____.

Answer: Spontaneous

Q4. The equation ΔG = ΔH – TΔS includes T representing _____.

Answer: Absolute temperature

Q5. The heat content of a system is called _____.

Answer: Enthalpy

Q6. The measure of randomness in a system is known as _____.

Answer: Entropy

Q7. Reactions that release heat energy are termed _____ reactions.

Answer: Exothermic

Q8. Reactions requiring absorption of heat are called _____ reactions.

Answer: Endothermic

Q9. The universal energy currency of the cell is _____.

Answer: ATP

Q10. ATP stores energy mainly in its _____ bonds.

Answer: Phosphoanhydride

Q11. Hydrolysis of ATP to ADP releases _____ energy.

Answer: Free

Q12. ATP generation directly coupled to metabolic reactions is called _____ phosphorylation.

Answer: Substrate-level

Q13. ATP synthesis linked with electron transport is called _____ phosphorylation.

Answer: Oxidative

Q14. The Electron Transport Chain is located in the _____ mitochondrial membrane.

Answer: Inner

Q15. The primary electron donors to the ETC are NADH and _____.

Answer: $FADH_2$

Q16. Oxygen serves as the final electron _____ in the ETC.

Answer: Acceptor

Q17. Reduction of oxygen at the end of the ETC produces _____.

Answer: Water

Q18. Complex I of the ETC is also known as NADH _____.

Answer: Dehydrogenase

Q19. Succinate dehydrogenase corresponds to ETC Complex _____.

Answer: II

Q20. Cytochrome oxidase is identified as ETC Complex _____.

Answer: IV

Q21. The only non-protein mobile electron carrier in the ETC is _____.

Answer: Coenzyme Q

Q22. The mobile carrier transferring electrons between Complex III and IV is _____.

Answer: Cytochrome c

Q23. Proton pumping across the inner mitochondrial membrane creates a proton _____.

Answer: Gradient

Q24. ATP production driven by proton movement is explained by the _____ theory.

Answer: Chemiosmotic

Q25. The flow of protons back into the mitochondrial matrix occurs through _____.

Answer: ATP synthase

Q26. ATP synthase consists of F_0 and F_1 _____.

Answer: Regions

Q27. The F_0 portion of ATP synthase functions as a proton _____.

Answer: Channel

Q28. The catalytic activity of ATP synthase resides in the _____ region.

Answer: F_1

Q29. ATP synthesis is catalyzed specifically by the β subunits of the _____ region.

Answer: F_1

Q30. The $\alpha_3\beta_3$ structure forms the hexameric ring of _____.

Answer: ATP synthase

Q31. Chemiosmosis requires an intact membrane, proton gradient, and _____.

Answer: ATP synthase

Q32. The energy required for ATP synthesis comes from the movement of _____.

Answer: Protons

Q33. Biological catalysts that accelerate reactions without being consumed are called _____.

Answer: Enzymes

Q34. Enzymes lower the _____ energy of reactions.

Answer: Activation

Q35. The specific region where substrates bind to enzymes is called the _____ site.

Answer: Active

Q36. Non-protein molecules required for enzyme activity are called _____.

Answer: Cofactors

Q37. Organic cofactors participating in enzymatic reactions are called _____.

Answer: Coenzymes

Q38. Vitamins often function as precursors of _____.

Answer: Coenzymes

Q39. Enzymes that differ in structure but catalyze the same reaction are called _____.

Answer: Isozymes

Q40. Hexokinase catalyzes the phosphorylation of _____.

Answer: Glucose

Q41. Glucokinase is predominantly found in the liver and pancreatic _____ cells.

Answer: Beta

Q42. Phosphofructokinase-1 catalyzes the formation of fructose _____.

Answer: 1,6-bisphosphate

Q43. High levels of ATP generally _____ phosphofructokinase-1 activity.

Answer: Inhibit

Q44. AMP acts as an _____ of phosphofructokinase-1.

Answer: Activator

Q45. Fructose 2,6-bisphosphate is a potent activator of _____.

Answer: PFK-1

Q46. The conversion of phosphoenolpyruvate to pyruvate is catalyzed by _____ kinase.

Answer: Pyruvate

Q47. Pyruvate kinase is activated by fructose _____.

Answer: 1,6-bisphosphate

Q48. During low blood glucose levels, pyruvate kinase is inactivated by _____.

Answer: Phosphorylation

Q49. The end product of glycolysis under aerobic conditions is _____.

Answer: Pyruvate

Q50. Glycolysis occurs in the _____ of the cell.

Answer: Cytosol

Q51. The irreversible step converting glucose to glucose-6-phosphate is catalyzed by _____.

Answer: Hexokinase

Q52. The rate-limiting enzyme of glycolysis is _____.

Answer: Phosphofructokinase-1

Q53. Citrate acts as an allosteric _____ of phosphofructokinase-1.

Answer: Inhibitor

Q54. High ATP levels signal a high-energy state and _____ glycolysis.

Answer: Inhibit

Q55. AMP stimulates glycolysis by activating _____.

Answer: PFK-1

Q56. Fructose 2,6-bisphosphate coordinates glycolysis and _____.

Answer: Gluconeogenesis

Q57. The enzyme catalyzing the final step of glycolysis is _____.

Answer: Pyruvate kinase

Q58. Pyruvate kinase converts phosphoenolpyruvate into _____.

Answer: Pyruvate

Q59. Feed-forward activation of pyruvate kinase is caused by fructose _____.

Answer: 1,6-bisphosphate

Q60. The conversion of pyruvate to acetyl-CoA occurs in the mitochondrial _____.

Answer: Matrix

Q61. The enzyme complex linking glycolysis with the TCA cycle is the _____ complex.

Answer: Pyruvate dehydrogenase

Q62. Pyruvate dehydrogenase converts pyruvate into acetyl-_____.

Answer: CoA

Q63. The pyruvate dehydrogenase reaction is considered _____.

Answer: Irreversible

Q64. Thiamine pyrophosphate is derived from vitamin _____.

Answer: B1

Q65. Lipoic acid functions as a coenzyme in the _____ complex.

Answer: Pyruvate dehydrogenase

Q66. NAD^+ accepts electrons during PDH activity to form _____.

Answer: NADH

Q67. High concentrations of acetyl-CoA _____ pyruvate dehydrogenase.

Answer: Inhibit

Q68. Pyruvate dehydrogenase is inactivated by _____.

Answer: Phosphorylation

Q69. The TCA cycle is also known as the _____ acid cycle.

Answer: Citric

Q70. The TCA cycle occurs in the mitochondrial _____.

Answer: Matrix

Q71. Acetyl-CoA combines with oxaloacetate to form _____.

Answer: Citrate

Q72. The enzyme catalyzing citrate formation is _____ synthase.

Answer: Citrate

Q73. Citrate synthase is inhibited by ATP, NADH, citrate, and _____.

Answer: Succinyl-CoA

Q74. The enzyme converting citrate to isocitrate is _____.

Answer: Aconitase

Q75. Isocitrate dehydrogenase catalyzes oxidative _____ of isocitrate.

Answer: Decarboxylation

Q76. Isocitrate dehydrogenase is activated by ADP and _____.

Answer: Calcium

Q77. α-Ketoglutarate dehydrogenase produces succinyl-_____.

Answer: CoA

Q78. α-Ketoglutarate dehydrogenase requires the same cofactors as the _____ complex.

Answer: Pyruvate dehydrogenase

Q79. Succinyl-CoA synthetase generates _____ by substrate-level phosphorylation.

Answer: GTP

Q80. Succinate dehydrogenase is also known as ETC Complex _____.

Answer: II

Q81. The oxidation of succinate produces _____.

Answer: $FADH_2$

Q82. Fumarase catalyzes the hydration of fumarate to _____.

Answer: Malate

Q83. Malate dehydrogenase converts malate into _____.

Answer: Oxaloacetate

Q84. One turn of the TCA cycle generates _____ molecules of NADH.

Answer: Three

Q85. Each acetyl-CoA oxidation in the TCA cycle yields _____ molecule of $FADH_2$.

Answer: One

Q86. The TCA cycle produces one molecule of _____ per turn.

Answer: GTP

Q87. Oxaloacetate functions primarily as a catalytic _____ in the TCA cycle.

Answer: Intermediate

Q88. Amphibolic pathways participate in both catabolic and _____ metabolism.

Answer: Anabolic

Q89. The TCA cycle is considered an _____ pathway.

Answer: Amphibolic

Q90. Intermediates removed from the TCA cycle are replenished through _____ reactions.

Answer: Anaplerotic

Q91. Pyruvate carboxylase converts pyruvate to _____.

Answer: Oxaloacetate

Q92. Pyruvate carboxylase requires _____ as a coenzyme.

Answer: Biotin

Q93. Acetyl-CoA activates _____ carboxylase.

Answer: Pyruvate

Q94. Electron transport is coupled with ATP synthesis through the _____ mechanism.

Answer: Chemiosmotic

Q95. ETC activity depends directly on the availability of _____.

Answer: Oxygen

Q96. Inhibition of ATP synthase eventually slows _____ transport.

Answer: Electron

Q97. Uncouplers increase oxygen consumption while decreasing ATP _____.

Answer: Synthesis

Q98. 2,4-Dinitrophenol acts as an _____ agent.

Answer: Uncoupling

Q99. Oligomycin inhibits the _____ channel of ATP synthase.

Answer: Proton

Q100. The major purpose of oxidative phosphorylation is generation of cellular _____.

Answer: ATP

Q101. The metabolic pathway responsible for glucose synthesis from non-carbohydrate precursors is called _____.

Answer: Gluconeogenesis

Q102. Gluconeogenesis occurs primarily in the liver and _____.

Answer: Kidneys

Q103. The first bypass enzyme of gluconeogenesis is _____ carboxylase.

Answer: Pyruvate

Q104. Pyruvate carboxylase converts pyruvate into _____.

Answer: Oxaloacetate

Q105. Biotin functions as a coenzyme for _____ carboxylase.

Answer: Pyruvate

Q106. Oxaloacetate is converted to phosphoenolpyruvate by PEP _____.

Answer: Carboxykinase

Q107. The enzyme fructose-1,6-bisphosphatase hydrolyzes fructose-1,6-bisphosphate to fructose-6-_____.

Answer: Phosphate

Q108. Fructose-1,6-bisphosphatase is inhibited by AMP and fructose _____.

Answer: 2,6-bisphosphate

Q109. Glucose-6-phosphatase converts glucose-6-phosphate into free _____.

Answer: Glucose

Q110. Glucose-6-phosphatase is present mainly in the liver and _____.

Answer: Kidneys

Q111. Lactate produced by muscles can serve as a substrate for _____.

Answer: Gluconeogenesis

Q112. The conversion of lactate to glucose between muscle and liver is called the _____ cycle.

Answer: Cori

Q113. Alanine participates in glucose synthesis through the glucose-_____ cycle.

Answer: Alanine

Q114. Gluconeogenesis requires energy in the form of ATP and _____.

Answer: GTP

Q115. Acetyl-CoA acts as an activator of pyruvate _____.

Answer: Carboxylase

Q116. Glycogen is the storage form of glucose in animals found mainly in liver and _____.

Answer: Muscle

Q117. Glycogenesis refers to the synthesis of _____.

Answer: Glycogen

Q118. Glycogenolysis is the breakdown of _____.

Answer: Glycogen

Q119. The activated glucose donor for glycogen synthesis is UDP-_____.

Answer: Glucose

Q120. Glycogen synthase catalyzes the formation of α-1,4 glycosidic _____.

Answer: Bonds

Q121. Glycogenin functions as a primer for glycogen _____.

Answer: Synthesis

Q122. Branching enzyme introduces α-1,6 glycosidic _____.

Answer: Bonds

Q123. Glycogen phosphorylase catalyzes glycogen _____.

Answer: Degradation

Q124. Glycogen phosphorylase releases glucose-1-_____.

Answer: Phosphate

Q125. Debranching enzyme is required for complete glycogen _____.

Answer: Breakdown

Q126. Phosphoglucomutase converts glucose-1-phosphate into glucose-6-_____.

Answer: Phosphate

Q127. Liver glycogen maintains blood glucose _____.

Answer: Levels

Q128. Muscle glycogen primarily provides energy for muscle _____.

Answer: Contraction

Q129. Insulin stimulates glycogen _____.

Answer: Synthesis

Q130. Glucagon promotes glycogen _____.

Answer: Breakdown

Q131. Epinephrine activates glycogenolysis during stress and _____.

Answer: Exercise

Q132. Glycogen synthase is activated by _____.

Answer: Dephosphorylation

Q133. Glycogen phosphorylase is activated by _____.

Answer: Phosphorylation

Q134. The Pentose Phosphate Pathway occurs in the cellular _____.

Answer: Cytosol

Q135. The major reducing equivalent produced by the Pentose Phosphate Pathway is _____.

Answer: NADPH

Q136. NADPH is required for reductive biosynthesis and antioxidant _____.

Answer: Defense

Q137. The rate-limiting enzyme of the Pentose Phosphate Pathway is glucose-6-phosphate _____.

Answer: Dehydrogenase

Q138. Glucose-6-phosphate dehydrogenase deficiency may cause hemolytic _____.

Answer: Anemia

Q139. Ribose-5-phosphate generated by the PPP is required for nucleotide _____.

Answer: Synthesis

Q140. The oxidative phase of the PPP generates NADPH and ribulose-5-_____.

Answer: Phosphate

Q141. Transketolase requires _____ pyrophosphate as a coenzyme.

Answer: Thiamine

Q142. Transaldolase participates in the non-oxidative phase of the _____.

Answer: PPP

Q143. Rapidly dividing cells require ribose-5-phosphate for nucleic acid _____.

Answer: Synthesis

Q144. NADPH maintains reduced glutathione through glutathione _____.

Answer: Reductase

Q145. Reduced glutathione protects cells against oxidative _____.

Answer: Damage

Q146. Red blood cells depend heavily on the Pentose Phosphate Pathway for NADPH_____.

Answer: Production

Q147. Insulin generally promotes anabolic pathways and nutrient _____.

Answer: Storage

Q148. Glucagon primarily functions to increase blood glucose _____.

Answer: Levels

Q149. Reciprocal regulation prevents simultaneous glycolysis and _____.

Answer: Gluconeogenesis

Q150. Fructose-2,6-bisphosphate coordinates the regulation of glycolysis and _____.

Answer: Gluconeogenesis

Q151. Fatty acid oxidation primarily occurs in the mitochondrial _____.

Answer: Matrix

Q152. The breakdown of fatty acids into acetyl-CoA units is called _____.

Answer: β-oxidation

Q153. Activation of fatty acids occurs in the cellular _____.

Answer: Cytosol

Q154. Fatty acids are activated to form fatty acyl-_____.

Answer: CoA

Q155. Transport of long-chain fatty acids into mitochondria requires _____.

Answer: Carnitine

Q156. Carnitine palmitoyltransferase I is located on the outer mitochondrial _____.

Answer: Membrane

Q157. Malonyl-CoA inhibits carnitine palmitoyltransferase _____.

Answer: I

Q158. The first step of β-oxidation is catalyzed by acyl-CoA _____.

Answer: Dehydrogenase

Q159. The initial oxidation step of β-oxidation produces _____.

Answer: $FADH_2$

Q160. Hydration of enoyl-CoA is catalyzed by _____.

Answer: Enoyl-CoA hydratase

Q161. The second oxidation step of β-oxidation generates _____.

Answer: NADH

Q162. Thiolase catalyzes the cleavage of β-ketoacyl-_____.

Answer: CoA

Q163. Each cycle of β-oxidation shortens the fatty acid by _____ carbon atoms.

Answer: Two

Q164. Palmitic acid contains _____ carbon atoms.

Answer: Sixteen

Q165. Complete oxidation of palmitate yields _____ molecules of acetyl-CoA.

Answer: Eight

Q166. Unsaturated fatty acid oxidation requires additional _____.

Answer: Enzymes

Q167. Odd-chain fatty acid oxidation produces propionyl-_____.

Answer: CoA

Q168. Propionyl-CoA is converted into succinyl-_____.

Answer: CoA

Q169. Propionyl-CoA carboxylase requires _____ as a coenzyme.

Answer: Biotin

Q170. Methylmalonyl-CoA mutase requires vitamin _____.

Answer: B12

Q171. Fatty acid synthesis occurs mainly in the cellular _____.

Answer: Cytosol

Q172. The precursor for fatty acid synthesis is acetyl-_____.

Answer: CoA

Q173. Citrate transports acetyl-CoA from mitochondria to the _____.

Answer: Cytosol

Q174. Acetyl-CoA carboxylase catalyzes the formation of malonyl-_____.

Answer: CoA

Q175. Acetyl-CoA carboxylase is the _____ enzyme of fatty acid synthesis.

Answer: Rate-limiting

Q176. Biotin serves as a coenzyme for acetyl-CoA _____.

Answer: Carboxylase

Q177. Citrate acts as an allosteric _____ of acetyl-CoA carboxylase.

Answer: Activator

Q178. Palmitoyl-CoA inhibits acetyl-CoA _____.

Answer: Carboxylase

Q179. Insulin promotes fatty acid synthesis by _____ acetyl-CoA carboxylase.

Answer: Activating

Q180. Fatty acid synthase is a multifunctional enzyme _____.

Answer: Complex

Q181. The reducing power for fatty acid synthesis is provided by _____.

Answer: NADPH

Q182. The primary product of fatty acid synthesis is _____.

Answer: Palmitate

Q183. Essential fatty acids cannot be synthesized by the human _____.

Answer: Body

Q184. Linoleic acid is classified as an _____ fatty acid.

Answer: Essential

Q185. Ketone bodies are synthesized in the liver _____.

Answer: Mitochondria

Q186. Ketogenesis increases during prolonged _____.

Answer: Fasting

Q187. The precursor for ketone body synthesis is acetyl-_____.

Answer: CoA

Q188. Acetoacetate is a primary _____ body.

Answer: Ketone

Q189. β-Hydroxybutyrate is formed by reduction of _____.

Answer: Acetoacetate

Q190. Acetone is produced by spontaneous decarboxylation of _____.

Answer: Acetoacetate

Q191. The liver cannot utilize ketone bodies due to absence of thiophorase _____.

Answer: Enzyme

Q192. Extrahepatic tissues convert ketone bodies into acetyl-_____.

Answer: CoA

Q193. Diabetic ketoacidosis is associated with uncontrolled _____.

Answer: Diabetes

Q194. Cholesterol synthesis primarily occurs in the _____.

Answer: Cytosol

Q195. The precursor molecule for cholesterol synthesis is acetyl-_____.

Answer: CoA

Q196. HMG-CoA reductase catalyzes the formation of _____.

Answer: Mevalonate

Q197. HMG-CoA reductase is the _____ enzyme of cholesterol synthesis.

Answer: Rate-limiting

Q198. Statin drugs inhibit HMG-CoA _____.

Answer: Reductase

Q199. Cholesterol serves as a precursor for steroid hormones, bile acids, and _____.

Answer: Vitamin D

Q200. LDL primarily functions to transport cholesterol to _____ tissues.

Answer: Peripheral

🚨 200 Most Important Repeated Conceptual MCQs 🚨

⚠ Important Note:

These MCQs have been compiled from repeated exam patterns, important concepts, and previous assessments to support your preparation.

📚 However, do NOT rely solely on this compilation.

For the best results, thoroughly study your Handouts, PPTs, Video Lectures, and Course Material as well.

Prepare Smart, Work Hard, and Aim for Excellence.

Best of Luck! 🍀📚

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